

Initial Project Planning Template

Date	30 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Input	USN-1	As a farmer, I can enter soil parameters (N, P, K, temperature, humidity, pH, rainfall) into the system.	3	High	CH.V.Rithesh Reddy	01 Jul 2026	07 Jul 2026
		USN-2	As a farmer, I can submit my data and receive an instant crop recommendation from the ML model.	3	High	CH.V.Rithesh Reddy	01 Jul 2026	07 Jul 2026
		USN-3	As a farmer, I can view the top 3 recommended crops ranked by suitability score.	2	High	D.Dhanush	01 Jul 2026	07 Jul 2026
Sprint-2	Crop Suitability	USN-4	As a user, I can select a specific crop and check if my current soil-climate conditions are suitable for it.	3	High	CH.V.Rithesh Reddy	08 Jul 2026	14 Jul 2026
		USN-5	As a user, I can view a compatibility report showing which parameters match or deviate from ideal crop ranges.	2	Medium	D.Dhanush	08 Jul 2026	14 Jul 2026
		USN-6	As a user, I receive suggestions on how to adjust soil parameters (e.g., add fertilizer) to improve suitability.	2	Medium	Tejandra Vasantha	08 Jul 2026	14 Jul 2026
Sprint-3	Resource Optimization	USN-7	As a farmer, I can enter my crop type and receive fertilizer quantity recommendations (N, P, K dosage).	3	High	D.Dhanush	15 Jul 2026	21 Jul 2026
		USN-8	As a farmer, I can view estimated cost savings when following the optimized fertilizer plan.	2	Medium	Tejandra Vasantha	15 Jul 2026	21 Jul 2026
Sprint-4	Analytics Dashboard	USN-9	As a researcher, I can view aggregate crop-environment pattern charts across multiple farm entries.	3	High	D.Dhanush	22 Jul 2026	28 Jul 2026
		USN-10	As a policymaker, I can filter analytics by region, season, and crop type to identify production trends.	2	Medium	Tejandra Vasantha	22 Jul 2026	28 Jul 2026
		USN-11	As an admin, I can export analytics reports as CSV or PDF for external sharing.	2	Low	CH.V.Rithesh Reddy	22 Jul 2026	28 Jul 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-5	User Interface	USN-12	As a farmer, I can use the system on a mobile-friendly web browser without installing any app.	2	High	Tejandra Vasantha	29 Jul 2026	04 Aug 2026
		USN-13	As a user, I can view the system in my preferred language (multilingual support).	2	Medium	Tejandra Vasantha	29 Jul 2026	04 Aug 2026
		USN-14	As a user, I can view a history of my past crop recommendations and yield outcomes.	2	Medium	D.Dhanush	29 Jul 2026	04 Aug 2026
Sprint-6	Testing & Deployment	USN-15	As a developer, I can run unit tests on the ML model to validate recommendation accuracy above 90%.	3	High	CH.V.Rithesh Reddy	05 Aug 2026	11 Aug 2026
		USN-16	As a developer, I can deploy the OptiCrop application to a cloud server (AWS/Heroku) for public access.	3	High	D.Dhanush	05 Aug 2026	11 Aug 2026

Sprint Velocity Summary

Sprint	Epic / Focus	Total Story Points	Start Date	End Date
Sprint-1	Data Input & Crop Recommendation	8	01 Jul 2026	07 Jul 2026
Sprint-2	Crop Suitability Assessment	7	08 Jul 2026	14 Jul 2026
Sprint-3	Fertilizer & Resource Optimization	5	15 Jul 2026	21 Jul 2026
Sprint-4	Analytics Dashboard	7	22 Jul 2026	28 Jul 2026
Sprint-5	User Interface & Experience	6	29 Jul 2026	04 Aug 2026
Sprint-6	Testing & Deployment	6	05 Aug 2026	11 Aug 2026
Total		39		